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Abstract

Aims

While elevated heart rate is an established marker of risk in HF, the prognostic relevance of heart rate is less certain in patients with comorbid atrial fibrillation/flutter (AFF). We investigated the associations between heart rate and outcomes according to AFF status in 5 HFmrEF/HFpEF clinical trials.

Methods

In a participant-level pooled analysis of the CHARM-Preserved, I-PRESERVE, TOPCAT-Americas, PARAGON-HF, and DELIVER trials, associations between baseline heart rate and outcomes according to AFF status on ECG at enrolment were assessed with multivariable Cox and Poisson regression models. The primary outcome was CV death or HF hospitalization.

Results

Among 19 975 participants, 5816 (29%) had AFF on baseline ECG. Patients with AFF were older, more frequently male, had a higher baseline heart rate (75 vs 68 bpm, $P < .001$), and had an increased risk for the primary outcome (adj HR 1.19 [1.10–1.27]). A significant interaction between heart rate, AFF status, and clinical outcomes was observed, such that patients in sinus rhythm had higher event rates with increasing heart rates, while the incident rates for participants in AFF were similar across the range of baseline heart rate ($P_{\text{interaction}} < .001$ for the primary outcome). This relationship was not further modified by concomitant β -blocker use.

Conclusions

In this analysis of five HFmrEF/HFpEF trials, baseline heart rate was associated with significantly higher rates of events only in patients in sinus rhythm but not in those with AFF. The optimal management of AFF in the context of HFmrEF/HFpEF requires a dedicated study.

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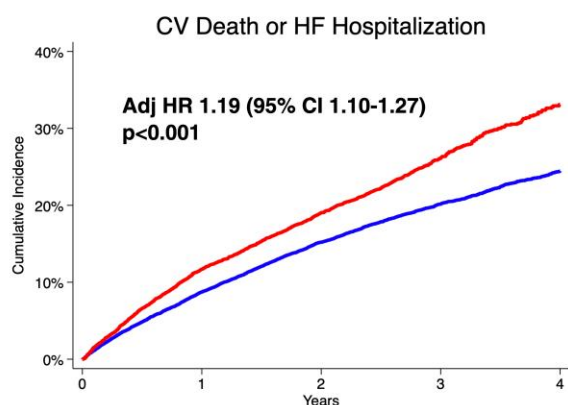
Graphical abstract

**Participant-level pooled analysis of 5 HFmrEF/HFpEF randomized clinical trials:
CHARM-Preserved I-PRESERVE TOPCAT-Americas PARAGON-HF DELIVER
(n=19,975)**

5,816 (29%) patients had atrial fibrillation/flutter on baseline ECG

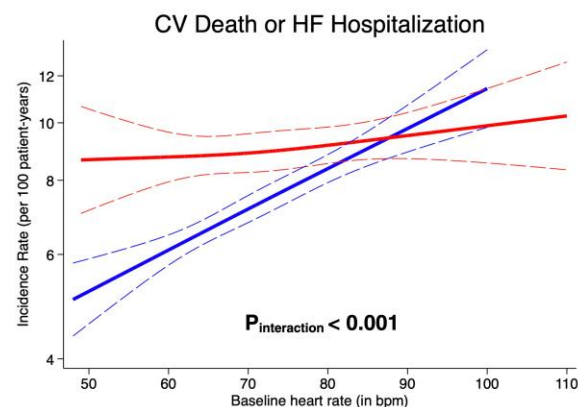


ATRIAL FIBRILLATION/FLUTTER AND OUTCOMES



— Sinus rhythm

BASELINE HEART RATE AND OUTCOMES



— Atrial fibrillation/flutter

In this analysis of nearly 20,000 patients with HFmrEF/HFpEF across 5 clinical trials, baseline heart rate was associated with poor outcomes only among participants enrolled in sinus rhythm while in patients presenting in atrial fibrillation/flutter the incident rates of events did not vary across baseline heart rate

Atrial fibrillation/flutter, heart rate, and clinical outcomes in HFmrEF/HFpEF clinical trials. Left panel: Kaplan-Meier curves showing the cumulative incidence of CV death or HF hospitalization for participants in sinus rhythm and with atrial fibrillation/flutter. Right panel: restricted cubic splines showing the association between baseline heart rate as a continuous variable and incident rates per 100 patient-years of CV death or HF hospitalization for participants in sinus rhythm (blue line) and with fibrillation/flutter (red line). CI, confidence interval; CV, cardiovascular; HF, heart failure; HFmrEF, heart failure with mildly reduced ejection fraction; HFpEF, heart failure with preserved ejection fraction; HR, hazard ratio.

Keywords

Atrial fibrillation • Clinical trials • Heart failure • Heart rate

Introduction

Resting heart rate is an established prognostic indicator in people with heart failure (HF). Multiple studies have consistently demonstrated the strong association between elevated heart rate and increased risk of clinical events in patients with heart failure and reduced ejection fraction (HFrEF) for whom current guidelines recommend targeting a heart rate of less than 70 bpm if in sinus rhythm.^{1,2} On the other hand, the optimal heart rate in the context of heart failure with mildly reduced or preserved ejection fraction (HFmrEF/HFpEF) is less certain.³⁻⁶

Heart rate management is especially relevant for patients with comorbid atrial fibrillation/flutter (AFF).⁷ While a rate control strategy is considered in many patients with HF and AFF, the ideal target heart

rate remains to be fully elucidated, as reflected by the weak recommendations in both American and European guidelines.^{2,8} A *post-hoc* analysis of the RACE-II (Rate Control Efficacy in permanent AF II) study on participants with atrial fibrillation and HF showed no significant difference between a strict vs a more lenient rate control therapy.⁹ Secondary analyses of clinical trials enrolling participants with HF showed similar observations, suggesting a weak association between heart rate and outcomes among people with concomitant AFF.^{4,10} However, most of this evidence has been derived from studies that included HFrEF patients, whereas the prognostic role of heart rate in those with HFpEF and AFF is less explored.

We conducted a participant-level pooled analysis of five randomized clinical trials (CHARM-preserved, I-PRESERVE, TOPCAT-Americas,

PARAGON-HF, and DELIVER) to investigate the association between baseline heart rate and clinical outcomes according to AFF status in patients with HFmrEF/HFpEF.

Methods

Study designs

The designs and primary results of CHARM-Preserved, I-PRESERVE, TOPCAT, PARAGON-HF, and DELIVER have been previously described and further details can be found in the [Supplement](#).^{11–15} All studies were global, multicenter, randomized clinical trials that enrolled participants with symptomatic HFmrEF/HFpEF. All participants provided informed written consent, and the study protocols were approved by the local ethics committees and institutional review boards at each participating site.

CHARM-Preserved was a placebo-controlled clinical trial testing the efficacy and safety of candesartan in participants aged ≥ 18 years who had a New York Heart Association function (NYHA) class II–IV for ≥ 4 weeks before randomization and a left ventricular ejection fraction (LVEF) $>40\%$.

I-PRESERVE was a placebo-controlled clinical trial testing the efficacy and safety of irbesartan in participants aged ≥ 60 years who had a NYHA functional class II–IV and an LVEF $\geq 45\%$.

TOPCAT was a placebo-controlled clinical trial testing the efficacy and safety of spironolactone in participants of ≥ 50 years of age, with an LVEF $\geq 45\%$ and a HF hospitalization within the previous 12 months or elevated natriuretic peptides. Patients with atrial fibrillation and a resting heart rate >90 bpm were excluded from the study. Due to the previously described regional heterogeneity in patient profiles, outcomes, treatment responses, and protocol adherence, only participants enrolled from North and South America (TOPCAT-Americas) were included in this analysis.¹⁶ Additionally, two patients were excluded because of missing baseline heart rates.

PARAGON-HF trial compared the efficacy and safety of sacubitril/valsartan vs valsartan alone in participants of ≥ 50 years of age who had a NYHA class II–IV, an LVEF $\geq 45\%$, evidence of structural heart disease, and elevated natriuretic peptides. Patients with AFF and a resting heart rate >110 bpm were excluded from the study.

DELIVER was a placebo-controlled clinical trial testing the efficacy and safety of dapagliflozin in participants aged ≥ 40 years who had a NYHA class II–IV, an LVEF $>40\%$, evidence of structural heart disease, and elevated natriuretic peptide levels.

The CHARM-preserved, I-PRESERVE, and DELIVER trials did not apply specific exclusion criteria regarding resting heart rate for either participants in sinus rhythm or those presenting with AFF.

AFF status and heart rate

AFF status was defined by the presence of either atrial fibrillation or atrial flutter documented on the randomization ECG. For all participants, resting heart rate was evaluated at baseline either during physical examination or from ECG tracings, and was divided into four categories: <60 bpm; ≥ 60 and <70 bpm; ≥ 70 and <80 bpm; ≥ 80 bpm. A more detailed description of heart rate in each trial can be found in the [Supplement](#).

Outcomes

The primary outcome in the present analysis was a composite of CV death or HF hospitalization (first event). The individual components of the primary outcome and all-cause mortality were additionally evaluated. All these events were independently adjudicated by clinical endpoint committees.

Statistical analysis

Baseline characteristics were summarized as median and interquartile range (IQR) for continuous variables and frequencies (%) for categorical variables. Baseline characteristics were compared using Student's *t*-tests and chi-square tests where appropriate. The distribution of baseline heart rate according to AFF status was visualized using kernel density estimation.

Event rates were estimated as the number of events over the total duration of follow-up and expressed as a rate per 100 patient-years.

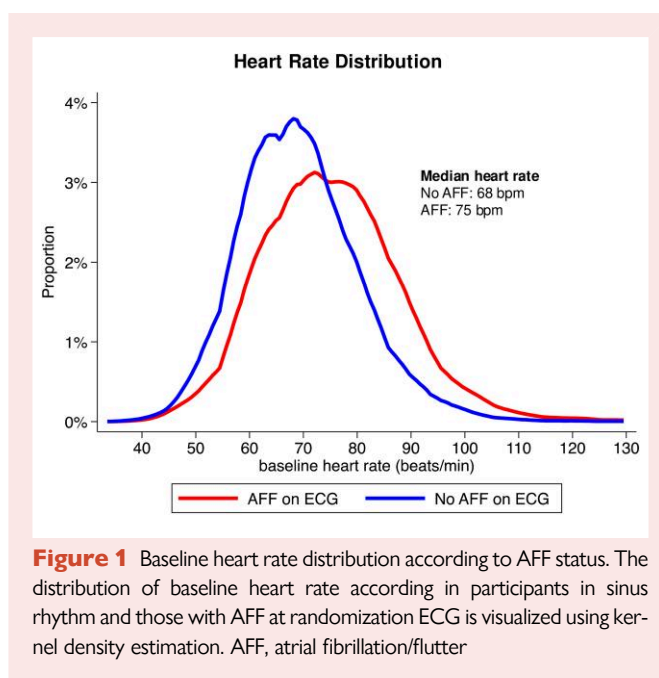


Figure 1 Baseline heart rate distribution according to AFF status. The distribution of baseline heart rate according in participants in sinus rhythm and those with AFF at randomization ECG is visualized using kernel density estimation. AFF, atrial fibrillation/flutter

Risk of the primary outcome, its individual components, and all-cause mortality were compared by AFF status using Cox proportional-hazards models—with and without adjustment for the following covariates: age, sex, race, diabetes, previous myocardial infarction, eGFR, NYHA class, systolic blood pressure, heart rate, LVEF, β -blockers, angiotensin-converting enzyme inhibitors/angiotensin receptor blockers (ACEi/ARB), diuretics, treatment, and body mass index (BMI). Time-to-event variables were further summarized using adjusted Kaplan–Meier cumulative incidence curves truncated at 4 years to account for heterogeneity in follow-up time across trials. The association between baseline heart rate as a categorical variable and the risk of clinical outcomes was analyzed with Cox proportional-hazards models adjusted for the aforementioned set of covariates plus digoxin and calcium channel blocker use at baseline, using as the reference category a heart rate <60 bpm. Two sensitivity analyses were conducted, excluding trials that had heart rate cut-offs for participants in AFF (TOPCAT and PARAGON-HF), and excluding participants who had a pacemaker at baseline. All Cox proportional-hazards models were stratified by trial. The association between baseline heart rate as a continuous variable, AFF status, and incident rates of clinical outcomes was evaluated with Poisson regression models using restricted cubic splines with three knots adjusted for the same set of covariates. All the analyses were conducted in the overall study population and also analysed separately for participants with HFmrEF (defined as an LVEF $>40\%$ and $<50\%$) and those with HFpEF (with an LVEF $\geq 50\%$).

Statistical analyses were performed using STATA version 18 (StataCorp, College Station, TX, USA). *P*-values of $<.05$ were considered statistically significant.

Results

Baseline characteristics according to AFF status

Among the pooled study population of 19 975 participants across five clinical trials, 5816 (29%) patients were in AFF at baseline ECG. The prevalence of AFF stratified by trial varied from 45% in DELIVER to 8% in CHARM-preserved and TOPCAT-Americas.

The median baseline heart rate in the overall study population was 70 bpm (interquartile range 62–78 bpm, min 35 bpm, max 162 bpm). The distribution of heart rate according to AFF status is displayed in [Figure 1](#). Participants with AFF had a higher median baseline heart

Table 1 Baseline characteristics according to heart rate categories in participants in sinus rhythm at baseline ECG

	HR < 60 bpm n = 2198	HR ≥ 60 & <70 bpm n = 5303	HR ≥ 70 & <80 bpm n = 4181	HR ≥ 80 bpm n = 2477	P for trend
Age—years	72 [65, 78]	71 [65, 77]	70 [64, 77]	69 [62, 76]	<.001
Men	1189 (54.1%)	2674 (50.4%)	1941 (46.4%)	1183 (47.8%)	<.001
Race					
White	1793 (81.6%)	4450 (83.9%)	3453 (82.6%)	1960 (79.1%)	.009
Black	97 (4.4%)	214 (4.0%)	175 (4.2%)	178 (7.2%)	
Asian	188 (8.6%)	423 (8.0%)	353 (8.4%)	235 (9.5%)	
Other	120 (5.5%)	216 (4.1%)	200 (4.8%)	104 (4.2%)	
Trial					
CHARM-Preserved (enrolment 1999–2000)	371 (16.9%)	867 (16.3%)	712 (17.0%)	595 (24.0%)	.30
I-PRESERVE (enrolment 2001–2003)	302 (13.7%)	1268 (23.9%)	1185 (28.3%)	676 (27.3%)	
TOPCAT-Americas (enrolment 2006–2012)	249 (11.3%)	484 (9.1%)	362 (8.7%)	225 (9.1%)	
PARAGON-HF (enrolment 2014–2016)	660 (30.0%)	1213 (22.9%)	880 (21.0%)	491 (19.8%)	
DELIVER (enrolment 2018–2020)	616 (28.0%)	1471 (27.7%)	1042 (24.9%)	490 (19.8%)	
Randomized treatment	1094 (49.8%)	2632 (49.6%)	2090 (50.0%)	1280 (51.7%)	.16
Medical history					
Hypertension	1896 (86.3%)	4623 (87.2%)	3649 (87.3%)	2078 (83.9%)	.021
Diabetes	724 (32.9%)	1986 (37.5%)	1726 (41.3%)	1103 (44.5%)	<.001
Myocardial infarction	755 (34.3%)	1790 (33.8%)	1277 (30.5%)	673 (27.2%)	<.001
Stroke	197 (9.0%)	467 (8.8%)	366 (8.8%)	232 (9.4%)	.65
Body mass index—kg/m ²	29 [26, 33]	29 [26, 33]	29 [26, 33]	30 [26, 34]	<.001
eGFR—ml/min/1.73 m ²	62 [48, 76]	64 [50, 79]	66 [51, 81]	66 [52, 85]	<.001
NYHA class					
I	37 (1.7%)	64 (1.2%)	49 (1.2%)	33 (1.3%)	<.001
II	1505 (68.5%)	3193 (60.2%)	2390 (57.2%)	1319 (53.3%)	
III	642 (29.2%)	2007 (37.9%)	1689 (40.4%)	1078 (43.6%)	
IV	14 (0.6%)	37 (0.7%)	51 (1.2%)	44 (1.8%)	
HFmrEF	469 (21.3%)	1226 (23.1%)	992 (23.7%)	605 (24.4%)	.012
HFpEF	1728 (78.6%)	4075 (76.8%)	3185 (76.2%)	1869 (75.5%)	.010
Left ventricular ejection fraction—%	57 [50, 64]	55 [50, 62]	55 [50, 62]	55 [50, 62]	<.001
Systolic blood pressure—mmHg	132 [120, 143]	131 [120, 142]	132 [122, 143]	134 [122, 146]	<.001
Diastolic blood pressure—mmHg	71 [64, 80]	75 [68, 80]	79 [70, 83]	80 [70, 85]	<.001
Background medications					
Beta blockers	1799 (82.0%)	4041 (76.2%)	2807 (67.2%)	1368 (55.3%)	<.001
ACEi/ARB	1638 (74.5%)	3895 (73.5%)	2962 (70.8%)	1724 (69.6%)	<.001
Diuretics	1921 (87.6%)	4654 (87.8%)	3621 (86.7%)	2163 (87.4%)	.41
Calcium channel blockers	765 (34.9%)	1932 (36.4%)	1455 (34.8%)	898 (36.3%)	.84
Digoxin	80 (3.6%)	284 (5.4%)	285 (6.8%)	220 (8.9%)	.001
Oral anticoagulants	491 (22.4%)	997 (18.8%)	652 (15.6%)	407 (16.4%)	.001
Pacemaker	67 (3.0%)	515 (9.7%)	400 (9.6%)	203 (8.2%)	.001

ACEi, angiotensin-converting enzyme inhibitors; ARB, angiotensin receptor blockers; eGFR, estimated glomerular filtration rate; HFmrEF, heart failure with mildly reduced ejection fraction; HFpEF, heart failure with preserved ejection fraction; HR, heart rate; IQR, interquartile range; NYHA, New York Heart Association.

rate compared to those in sinus rhythm (75 vs 68 bpm, $P < .001$). Participants from trials without heart rate cut-offs as exclusion criteria (CHARM-preserved, I-PRESERVE, and DELIVER) had a similar baseline heart rate: median 70 bpm (75 bpm in patients with AFF vs 69 bpm in those in sinus rhythm), interquartile range 64–78 bpm, min 37 bpm, max 162 bpm.

Baseline characteristics according to heart rate categories for participants in sinus rhythm and those in AFF are summarized in [Tables 1](#) and [2](#). For both subgroups, participants with higher resting heart rates were

younger and more likely to be female. These persons were less likely to have had a prior myocardial infarction and among those in sinus rhythm, higher heart rates were associated with a progressively greater prevalence of diabetes mellitus. Overall, participants enrolled with higher resting heart rates had a worse NYHA functional class and a lower baseline LVEF. Regarding background medications, patients in sinus rhythm with lower heart rates were more likely to be treated with β -blockers, whereas digoxin use was more common at higher heart rates, irrespective of AFF status.

Table 2 Baseline characteristics according to heart rate categories in participants with atrial fibrillation/flutter at baseline ECG

	HR < 60 bpm n = 458	HR ≥ 60 & < 70 bpm n = 1509	HR ≥ 70 & < 80 bpm n = 1757	HR ≥ 80 bpm n = 2092	P for trend
Age—years	75 [70, 80]	75 [69, 80]	74 [68, 79]	73 [68, 79]	<.001
Men	306 (66.8%)	887 (58.8%)	949 (54.0%)	1035 (49.5%)	<.001
Race					
White	350 (76.4%)	1241 (82.2%)	1424 (81.0%)	1683 (80.4%)	.80
Black	20 (4.4%)	32 (2.1%)	29 (1.7%)	26 (1.2%)	
Asian	67 (14.6%)	183 (12.1%)	241 (13.7%)	316 (15.1%)	
Other	21 (4.6%)	53 (3.5%)	63 (3.6%)	67 (3.2%)	
Trial					
CHARM-Preserved (enrollment 1999–2000)	36 (7.9%)	118 (7.8%)	123 (7.0%)	201 (9.6%)	.92
I-PRESERVE (enrolment 2001–2003)	36 (7.9%)	159 (10.5%)	225 (12.8%)	277 (13.2%)	
TOPCAT-Americas (enrolment 2006–2012)	52 (11.4%)	161 (10.7%)	123 (7.0%)	109 (5.2%)	
PARAGON-HF (enrolment 2014–2016)	140 (30.6%)	396 (26.2%)	420 (23.9%)	596 (28.5%)	
DELIVER (enrolment 2018–2020)	616 (28.0%)	1471 (27.7%)	1042 (24.9%)	490 (19.8%)	
Randomized treatment	250 (54.6%)	745 (49.4%)	882 (50.2%)	1030 (49.2%)	.19
Medical history					
Hypertension	(1) 413 (90.2%)	1324 (87.7%)	1531 (87.1%)	1803 (86.2%)	.021
Diabetes	(1) 169 (36.9%)	533 (35.3%)	653 (37.2%)	752 (35.9%)	.96
Myocardial infarction	(1) 77 (16.8%)	262 (17.4%)	273 (15.5%)	283 (13.5%)	.002
Stroke	(1) 51 (11.2%)	177 (11.7%)	214 (12.2%)	226 (10.8%)	.52
Body mass index—kg/m ²	29 [26, 33]	29 [26, 34]	29 [26, 33]	30 [26, 34]	.06
eGFR—ml/min/1.73 m ²	59 [48, 72]	60 [48, 74]	60 [48, 74]	62 [49, 75]	.026
NYHA class					
I	8 (1.7%)	19 (1.3%)	14 (0.8%)	13 (0.6%)	.047
II	313 (68.3%)	970 (64.3%)	1135 (64.6%)	1343 (64.2%)	
III	136 (29.7%)	500 (33.1%)	596 (33.9%)	708 (33.9%)	
IV	1 (0.2%)	20 (1.3%)	12 (0.7%)	27 (1.3%)	
HFmrEF	87 (19.0%)	332 (22.0%)	382 (21.7%)	513 (24.5%)	.008
HFpEF	371 (81.0%)	1177 (78.0%)	1373 (78.1%)	1577 (75.4%)	.006
Left ventricular ejection fraction—%	57 [50, 62]	55 [50, 61]	55 [50, 61]	55 [50, 60]	<.001
Systolic blood pressure—mmHg	129 [117, 140]	127 [117, 138]	127 [118, 138]	128 [117, 140]	.41
Diastolic blood pressure—mmHg	70 [63, 79]	74 [67, 80]	76 [70, 81]	80 [70, 85]	<.001
Background medications					
Beta blockers	320 (69.9%)	1134 (75.1%)	1372 (78.1%)	1654 (79.1%)	<.001
ACEi/ARB	353 (77.1%)	1107 (73.4%)	1257 (71.5%)	1511 (72.2%)	.07
Diuretics	447 (97.6%)	1448 (96.0%)	1696 (96.5%)	2006 (95.9%)	.27
Calcium channel blockers	178 (38.9%)	478 (31.7%)	522 (29.7%)	589 (28.2%)	<.001
Digoxin	80 (17.5%)	321 (21.3%)	347 (19.7%)	532 (25.4%)	<.001
Oral anticoagulants	359 (78.4%)	1176 (77.9%)	1405 (80.0%)	1576 (75.3%)	.05
Pacemaker	27 (5.9%)	272 (18.0%)	217 (12.4%)	135 (6.5%)	<.001

ACEi, angiotensin-converting enzyme inhibitors; ARB, angiotensin receptor blockers; eGFR, estimated glomerular filtration rate; HFmrEF, heart failure with mildly reduced ejection fraction; HFpEF, heart failure with preserved ejection fraction; HR, heart rate; IQR, interquartile range; NYHA, New York Heart Association.

Baseline characteristics according to AFF status, independent of resting heart rate, are reported in [Supplementary Table S1](#).

Outcomes according to AFF status

Over a median follow-up of 33 months, 4402 primary events were observed; 2084 participants died due to CV causes and 3160 persons were hospitalized for HF. Overall, 3461 participants died from any cause. Clinical outcomes in participants based on AFF status are

reported in [Supplementary Table S2](#). Participants in AFF experienced the primary composite outcome of CV death or HF hospitalization with an incident rate of 10.3 per 100 patient-years compared to 7.4 per 100 patient-years in non-AFF group (adjusted HR 1.19 [95% CI 1.10–1.27]; $P < .001$), with consistent results observed for the individual components. Additionally, people presenting with AFF had a higher incident rate of all-cause mortality (7.5 vs 5.3 per 100 patient-years; adjusted HR 1.12 [95% CI 1.03–1.22]; $P = .005$; [Supplementary Figure S1](#)). The risk of adverse clinical outcomes related to AFF status appeared to

Table 3 Clinical outcomes in participants with and without atrial fibrillation/flutter according to baseline heart rate

	No AFF on ECG				AFF on ECG			
	No of events (rate per 100 py)	Hazard ratio (95% CI)	Adjusted hazard ratio ^a (95% CI)	P value	No of events (rate per 100 py)	Hazard ratio (95% CI)	Adjusted hazard ratio ^a (95% CI)	P value
CV death or HF hospitalization								
HR < 60 bpm	354 (5.8)	Reference	Reference	Reference	112 (10.1)	Reference	Reference	Reference
HR ≥ 60 & <70 bpm	1016 (6.8)	1.23 (1.09–1.38)	1.20 (1.05–1.36)	.006	379 (10.3)	1.02 (0.82–1.25)	1.08 (0.86–1.34)	.51
HR ≥ 70 & <80 bpm	926 (7.8)	1.45 (1.28–1.64)	1.45 (1.27–1.65)	<.001	438 (10.3)	1.02 (0.83–1.26)	1.08 (0.87–1.34)	.49
HR ≥ 80 bpm	656 (9.7)	1.79 (1.57–2.04)	1.72 (1.49–1.98)	<.001	521 (10.3)	1.01 (0.83–1.24)	1.16 (0.93–1.44)	.18
CV death								
HR < 60 bpm	149 (2.3)	Reference	Reference	Reference	47 (3.8)	Reference	Reference	Reference
HR ≥ 60 & <70 bpm	474 (2.9)	1.28 (1.06–1.54)	1.27 (1.04–1.55)	.017	184 (4.5)	1.15 (0.84–1.59)	1.17 (0.84–1.62)	.36
HR ≥ 70 & <80 bpm	450 (3.5)	1.53 (1.27–1.84)	1.57 (1.28–1.92)	<.001	218 (4.6)	1.18 (0.86–1.62)	1.21 (0.87–1.68)	.26
HR ≥ 80 bpm	327 (4.4)	1.90 (1.56–2.31)	1.96 (1.59–2.43)	<.001	235 (4.1)	1.06 (0.78–1.45)	1.19 (0.85–1.64)	.33
HF hospitalization								
HR < 60 bpm	257 (4.2)	Reference	Reference	Reference	92 (8.3)	Reference	Reference	Reference
HR ≥ 60 & <70 bpm	724 (4.8)	1.23 (1.07–1.42)	1.18 (1.02–1.37)	.030	280 (7.6)	0.92 (0.73–1.17)	1.00 (0.78–1.28)	.98
HR ≥ 70 & <80 bpm	650 (5.5)	1.45 (1.25–1.68)	1.42 (1.22–1.66)	<.001	310 (7.3)	0.89 (0.71–1.13)	0.95 (0.74–1.21)	.67
HR ≥ 80 bpm	454 (6.7)	1.75 (1.50–2.04)	1.60 (1.36–1.90)	<.001	393 (7.8)	0.94 (0.75–1.18)	1.08 (0.85–1.38)	.53
All-cause death								
HR < 60 bpm	257 (4.0)	Reference	Reference	Reference	82 (6.7)	Reference	Reference	Reference
HR ≥ 60 & <70 bpm	813 (5.1)	1.29 (1.12–1.49)	1.30 (1.12–1.51)	.001	321 (7.9)	1.17 (0.91–1.49)	1.19 (0.93–1.52)	.17
HR ≥ 70 & <80 bpm	715 (5.7)	1.45 (1.26–1.67)	1.50 (1.29–1.75)	<.001	359 (7.6)	1.12 (0.88–1.43)	1.17 (0.92–1.50)	.20
HR ≥ 80 bpm	493 (6.6)	1.75 (1.50–2.04)	1.83 (1.55–2.15)	<.001	421 (7.4)	1.11 (0.88–1.41)	1.26 (0.99–1.61)	.06

All analyses were stratified for trial.

ACEi, angiotensin-converting enzyme inhibitors; AFF, atrial fibrillation/flutter; ARB, angiotensin receptor blockers; BMI, body mass index; CI, confidence interval; CV, cardiovascular; eGFR, estimated glomerular filtration rate; HF, heart failure; HR, heart rate; LVEF, left ventricular ejection fraction; NYHA, New York Heart Association; py, patient-years.

^aAdjusted for: age, sex, race, diabetes, previous myocardial infarction, eGFR, NYHA class, systolic blood pressure, LVEF, beta blockers, digoxin, calcium channel blockers, ACEi/ARB, diuretics, treatment, BMI.

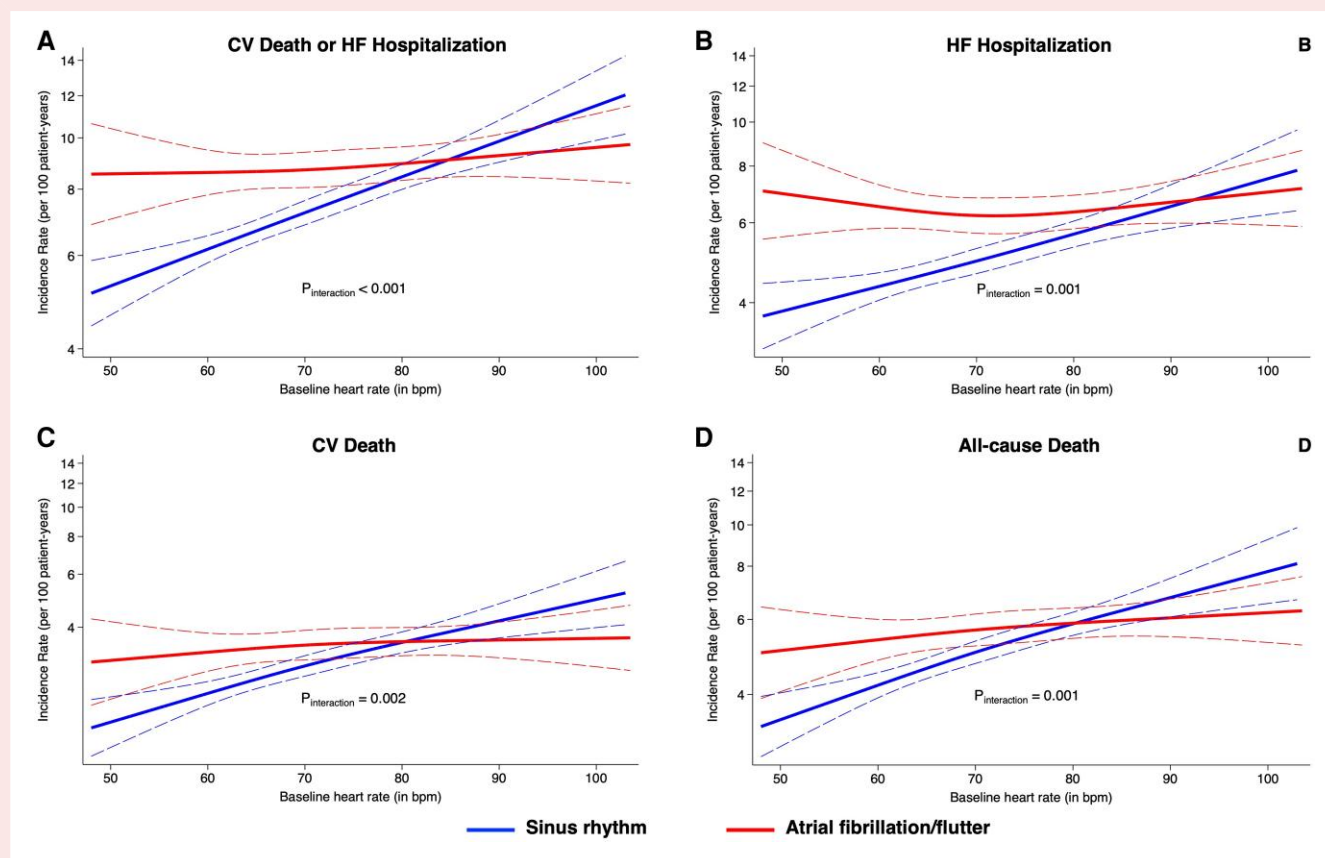


Figure 2 Baseline heart rate and incident rates of clinical outcomes according to AFF status. Restricted cubic splines showing the association between baseline heart rate as a continuous variable (x axis) and incident rates per 100 patient-years (y axis) of CV death or HF hospitalization (A), HF hospitalization (B), CV death (C), and all-cause mortality (D) for participants in sinus rhythm and with AFF. All models were stratified by trial and adjusted for: age, sex, race, diabetes, previous myocardial infarction, estimated glomerular filtration rate, New York Heart Association class, systolic blood pressure, left ventricular ejection fraction, β -blockers, digoxin, calcium channel blockers, angiotensin-converting enzyme inhibitors/angiotensin receptor blockers, diuretics, treatment, and body mass index. AFF, atrial fibrillation/flutter; CV, cardiovascular; HF, heart failure

be greater in participants with HFpEF compared to those with HFmrEF (Supplementary Table S2).

Baseline heart rate, AFF status, and clinical outcomes

In the overall study population, a higher baseline heart rate was associated with a higher risk of the primary outcome (per 10 bpm increase: adjusted HR 1.12 [95% CI 1.09–1.15; $P < .001$]), its individual components, and all-cause mortality (per 10 bpm increase: adjusted HR 1.12 [95% CI 1.08–1.15; $P < .001$]).

Clinical outcomes according to AFF status and heart rate as a categorical variable (reference category baseline heart rate <60 bpm) are reported in Table 3. In participants presenting in sinus rhythm, a baseline heart rate higher than the reference was associated with a greater risk for the primary outcome and all-cause mortality. Among participants presenting in AFF at randomization, no differences were observed in the risk of the primary composite outcome or its individual components across heart rate categories. A trend toward an increased risk of all-cause mortality was observed for people with a baseline heart rate ≥ 80 bpm (Table 3).

Consistent results were found after excluding participants who had a pacemaker at baseline (Supplementary Table S3), and when analysing only patients from trials without heart rate cut-offs (CHARM-preserved, I-PRESERVE, and DELIVER; Supplementary Table S4).

When baseline heart rate was analyzed as a continuous variable, a significant interaction with AFF status was observed for the risk of both the primary outcome and all-cause death. As shown in Figure 2, the incident rates of CV death or HF hospitalization in participants presenting in sinus rhythm increased with higher baseline heart rates ($P_{\text{interaction}} < .001$), with consistent findings when investigating the individual components and all-cause mortality ($P_{\text{interaction}} = .002$). Among persons with AFF at randomization, the incident rates of adverse clinical outcomes were similar across the range of baseline heart rate. The strength of the interaction among AFF status, heart rate, and event rates appeared to be more evident in HFpEF compared with HFmrEF (Figure 3, Supplementary Figure S2).

A three-way interaction test between AFF status, heart rate, and β -blocker use was performed, showing no statistically significant results ($P_{\text{interaction}} = .29$ for the primary outcome; $P_{\text{interaction}} = .94$ for all-cause mortality) suggesting that β -blocker use did not further modify the relationship between AFF, heart rate, and outcomes.

Discussion

In this participant-level pooled analysis of nearly 20 000 patients with HFmrEF/HFpEF across five clinical trials, resting heart rate was a strong independent predictor of adverse clinical outcomes only in those in sinus rhythm and not in those presenting in AFF (Graphical Abstract).

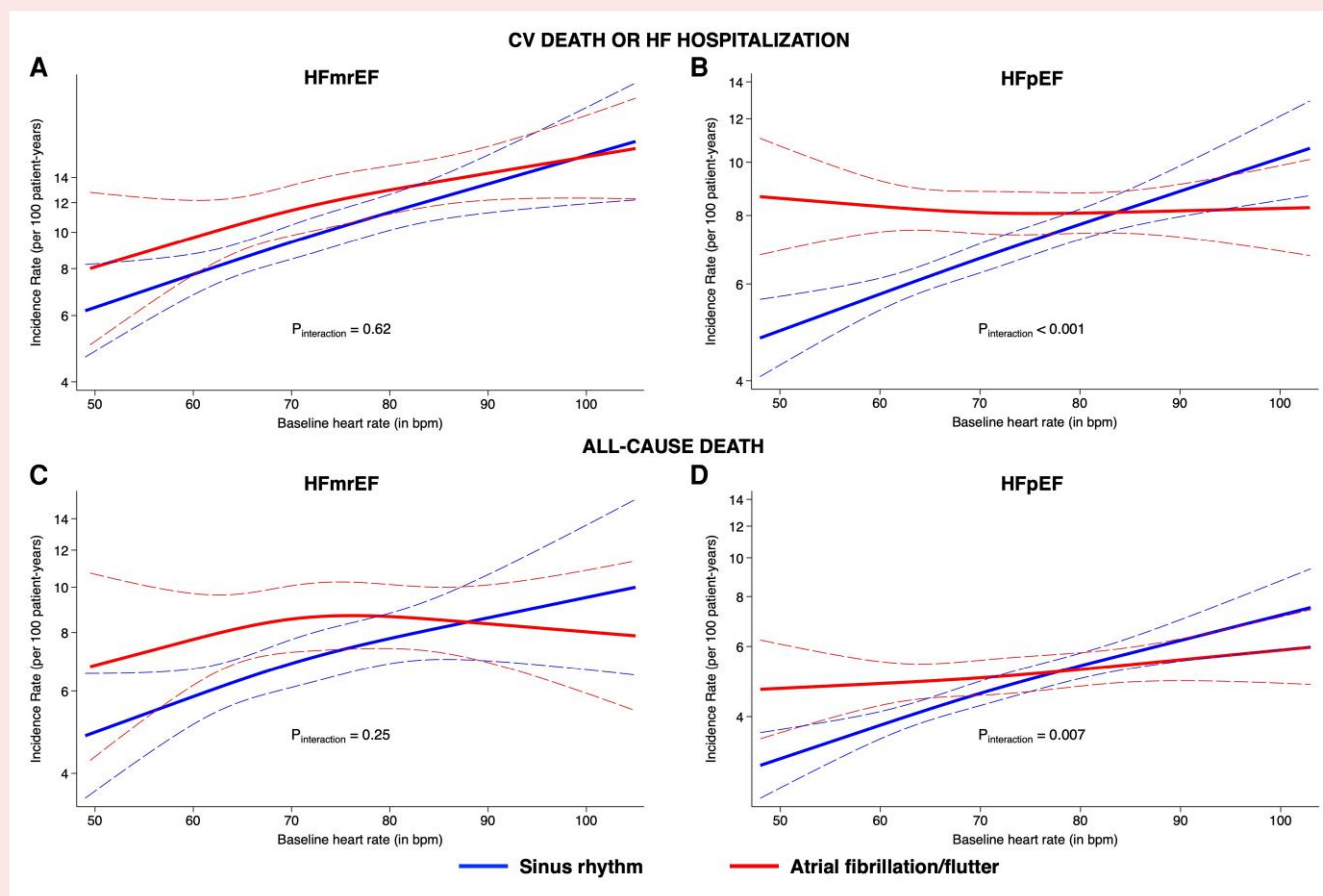


Figure 3 Baseline heart rate and incident rates of the primary outcome and all-cause death according to AFF status in participants with HFmrEF and HFpEF. Restricted cubic splines showing the association between baseline heart rate as a continuous variable (x axis) and incident rates per 100 patient-years (y axis) of CV death or HF hospitalization in HFmrEF (A) and in HFpEF (B), and all-cause mortality in HFmrEF (C) and HFpEF (D) for participants in sinus rhythm and with AFF. All models were adjusted for age, sex, race, trial, diabetes, previous myocardial infarction, estimated glomerular filtration rate, New York Heart Association class, systolic blood pressure, left ventricular ejection fraction, β -blockers, digoxin, calcium channel blockers, angiotensin-converting enzyme inhibitors/angiotensin receptor blockers, diuretics, treatment, and body mass index. AFF, atrial fibrillation/flutter; CV, cardiovascular; HF, heart failure; HFmrEF, heart failure with mildly reduced ejection fraction; HFpEF, heart failure with preserved ejection fraction

These observations question an aggressive rate control strategy for patients with HFmrEF/HFpEF with concomitant AFF and underscore the need for dedicated studies investigating the optimal management of persons with HFmrEF/HFpEF and AFF.

AFF has emerged as a dominant comorbid condition in patients with HFmrEF/HFpEF. The prevalence of AFF varied substantially across trials, with estimates from contemporary studies aligning with real-world data.⁷ Although persons with AFF enrolled in clinical trials tend to have a well-controlled ventricular rate due to the stringent follow-up visits and eligibility criteria, we confirmed that AFF is independently associated with a greater risk of events, in keeping with previous literature.^{17,18}

Higher resting heart rates are associated with fatal and nonfatal events among HFrEF patients in sinus rhythm.^{19,20} The extent to which this relationship also applies to patients with higher LVEFs is less investigated. Indeed, the pathophysiology underlying heart rate increase within the HF population varies according to the phenotype. In people with HFrEF, higher heart rates may represent a compensatory mechanism secondary to reduced stroke volume in an effort to maintain adequate organ perfusion.²¹ In the context of HFmrEF/HFpEF, multiple mechanisms may be involved, including chronotropic incompetence related to sinoatrial node dysfunction and enhanced sympathetic tone at rest.²² Despite differences in pathophysiology, a linear relationship

between higher heart rates and greater risk of events appears to be evident across the full spectrum of LVEF, confirming the clinical relevance of resting heart rate as a prognostic marker for HF people in sinus rhythm.²³ Nevertheless, dedicated research is required to conclusively determine the target heart rate range among people with HFmrEF/HFpEF and sinus rhythm.

The association among heart rate, AFF, and clinical outcomes in patients with HF is less investigated. A secondary analysis of two HFrEF clinical trials found no prognostic role of baseline heart rate among participants with AFF at enrolment, and similar observations were subsequently described also in persons with higher LVEF values.^{4,10,24} Our analysis of five HFmrEF/HFpEF clinical trials that enrolled participants over the last 20 years, confirms that incident rates of fatal and nonfatal events among participants enrolled with AFF were similar across the range of heart rate. This discrepancy with individuals in sinus rhythm raises intriguing pathophysiological questions about the mechanisms underlying heart rate variation in the context of HFpEF with concomitant AFF. First, ventricular response in this setting may be confounded on one hand by increased sympathetic tone, and on the other hand by impaired atrioventricular conduction, especially in patients with more advanced HF.²⁵ Second, compared to the general HFpEF population, this subgroup with AFF has distinct cardiac structural changes, including

larger left atrial volumes, increased LV stiffness, and impaired diastolic compliance.²⁶ Interestingly, recent evidence has suggested the potential benefit of higher resting heart rates in people with HFpEF, with a favourable haemodynamic impact in terms of reduced left ventricular filling pressures.²⁷ Therefore, it appears plausible that in the context of an exaggerated diastolic dysfunction caused by the coexistence of HFmrEF/HFpEF and AFF, an elevated heart rate may serve as an adaptive mechanism to facilitate LV unloading and relaxation. Indeed, these pathophysiological adaptations are typically more pronounced at higher LVEF values, which may explain why we observed an interaction between heart rate, prognosis, and rhythm status, particularly in the subgroup of participants with HFpEF.

From a therapeutic perspective, our findings align with previous studies suggesting that strict rate control in patients with AFF and HF may not be associated with clinical benefits.⁹ The clinical effects of β -blockers in patients with HF and AFF have been questioned, and β -blocker withdrawal in select people with HFpEF and chronotropic incompetence has been associated with improved functional capacity.^{28–30} As such, the management of these patients remains to be fully elucidated, and current guidelines underline the importance of a tailored therapy based on patients' symptoms and haemodynamic profile.^{1,8} Additionally, clinical decision-making may become even more challenging within the HFmrEF/HFpEF spectrum, which is highly heterogeneous and often requires a strict management of multiple comorbidities. Dedicated clinical trials are required to define optimal treatment strategies for HFmrEF/HFpEF and comorbid AFF.

Limitations

Our observations should be interpreted considering some limitations. First, this was a *post-hoc* analysis of clinical trials and therefore, our results should be considered hypothesis-generating. Second, participants were enrolled following stringent inclusion and exclusion criteria and therefore, our findings may not be generalizable to the heterogeneous HFmrEF/HFpEF population. Patients with AFF included in the five trials had a relatively well-controlled ventricular rate; therefore, based on our analysis, we cannot exclude an association between higher heart rates (>110 bpm) and risk of events in the HFmrEF/HFpEF population with concomitant AFF. Additionally, granular data regarding the type of AFF (paroxysmal, persistent or permanent) were not available and thus not analysed. However, since AFF status was determined by baseline ECG, most cases were persistent or permanent AFF. Third, the median follow-up time differed across trials, potentially affecting the absolute numbers of events. Fourth, distinct from the other trials, TOPCAT and PARAGON-HF had heart rate cut-offs for participants in AFF, possibly resulting in heterogeneity between trials. Nevertheless, all models were stratified by trial to account for variations in study designs, and the sensitivity analysis excluding TOPCAT-Americas and PARAGON-HF participants showed consistent results. Fifth, due to data availability related to the study designs, we analyzed heart rates only at baseline, while the longitudinal changes in heart rate during follow-up were not evaluated. Sixth, in the main analysis, we included participants with a pacemaker who may have a non-physiological heart rate, potentially confounding the associations between heart rate and outcomes. Nevertheless, a sensitivity analysis excluding patients with pacemakers showed consistent results. Finally, all our models were not adjusted for natriuretic peptides levels because they were not available in the CHARM-Preserved trial and in most TOPCAT-Americas participants.

Conclusions

In this analysis of nearly 20 000 patients with HFmrEF/HFpEF across five clinical trials, AFF was associated with a greater risk of fatal and nonfatal

cardiovascular events. Baseline heart rate was an independent predictor of poor outcome only among participants enrolled in sinus rhythm, while in persons presenting in AFF, the incident rates of events did not vary across the range of heart rate. These observations underscore the need for dedicated trials to identify the optimal management for patients with HFmrEF/HFpEF and concomitant AFF.

Supplementary data

Supplementary data are available at [European Journal of Heart Failure](https://academic.oup.com/eurjhf/article/28/2/206/8444741) online.

Declarations

Disclosure of Interest

Foà has nothing to disclose. Lu has received research grant support or served on advisory boards for Bayer, AstraZeneca, Boehringer Ingelheim, Cytokinetics, and Abbott. Vaduganathan has received research grant support, served on advisory boards, or had speaker engagements with Alnylam Pharmaceuticals, American Regent, Amgen, AstraZeneca, Bayer AG, Baxter Healthcare, BMS, Boehringer Ingelheim, Chiesi, Cytokinetics, Esperion, Fresenius Medical Care, Idorsia Pharmaceuticals, Lexicon Pharmaceuticals, Merck, Milestone Pharmaceuticals, Novartis, Novo Nordisk, Pharmacosmos, Recordati, Relypsa, Roche Diagnostics, Sanofi, and Tricog Health, and participates on clinical trial committees for studies sponsored by Amgen, AstraZeneca, Bayer AG, Boehringer Ingelheim, Galmed, Impulse Dynamics, Novartis, Occlutech, and Pharmacosmos. Claggett has received personal consulting fees from Alnylam, Bristol Myers Squibb, Cardior, Cardurion, Corvia, CVRx, Eli Lilly, Intellia, and Rocket and has served on a data and safety monitoring board (DSMB) for Novo Nordisk. A.S.S. has received institutional research grants (to Brigham and Women's Hospital) from Abbott, Alnylam, AstraZeneca, Bayer, Novartis and Pfizer as well as personal consulting fees from Abbott, Alnylam, AstraZeneca, Bayer, Biofourmis, Boston Scientific, Medpace, Medtronic, Merck, Novartis, Parexel, Porter Health, Regeneron, River2Renal, Roche, Veristat, Verily and Zydus. Skali reports research support or consulting fees from ABT Associates, Astellas Pharma, Emmi Solutions, Hikma Pharmaceuticals. Desai reports institutional research grants from Abbott, Alnylam, AstraZeneca, Bayer, Novartis, and Pfizer; and personal consulting fees from Abbott, Alnylam, AstraZeneca, Bayer, Biofourmis, Boston Scientific, Medpace, Medtronic, Merck, Novartis, Parexel, Porter Health, Regeneron, River 2 Renal, Roche, Veristat, Verily, and Zydus. Jhund reports speakers' fees from AstraZeneca, ProAdWise Communications; advisory board fees from AstraZeneca; research funding from AstraZeneca, Boehringer Ingelheim, Analog Devices Inc, Roche Diagnostics. PSJ's employer the University of Glasgow has been remunerated for clinical trial work from AstraZeneca, Bayer AG, Novartis, Novo Nordisk, and is director of GCTP Ltd. Zile has received research funding from Novartis; and has served as a consultant for Novartis, Abbott, Boston Scientific, CVRx, EBR, Endotronix, Ironwood, Merck, Medtronic, and Myokardia V Wave. Rouleau has received grants and consulting fees from DMC participation for Bayer, AstraZeneca, BMS, and Novartis. Anand has received consulting fees from Amgen, ARCA Pharma, AstraZeneca, Boehringer Ingelheim, Boston Scientific, LivaNova, and Novartis. Zannad reports personal fees from 89Bio, Abbott, Acceleron, Applied Therapeutics, Bayer, Betagenon, Boehringer, Bristol Myers Squibb, CVRx, Cambrian, Cardior, Cerenon Pharmaceutical, Cellprothera, CEVA, Inventiva, KBP, Merck, NovoNordisk, Owkin, Otsuka, Roche Diagnostics, Northsea and USa2; having stock options at G3Pharmaceutical and equities at Cerenon, Cardiorrenal, Eshmoun Clinical Research; and being the founder of Cardiovascular Clinical Trialists. Pfeffer reports research grant support (via institution) from Lexicon, Novartis; consultant to Alexion, Alnylam, Apnimed, AstraZeneca, Boehringer Ingelheim and Eli Lilly Alliance,

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Data Availability

Data underlying the findings described in this manuscript may be obtained in accordance with the data sharing policy of each of the trial sponsors (outlined below).

AstraZeneca's data sharing policy is described at <https://www.astrazenecaclinicaltrials.com/our-transparency-commitments/>. Data for studies directly listed on Vivli can be requested through Vivli at www.vivli.org. Data for studies not listed on Vivli could be requested through Vivli at <https://vivli.org/members/enquiries-about-studies-not-listed-on-the-vivli-platform/>. AstraZeneca's Vivli member page is also available outlining further details: <https://vivli.org/ourmember/astrazeneca/>.

Bristol-Myers Squibb's data sharing policy is detailed at <https://www.bms.com/researchers-and-partners/independent-research/data-sharing-request-process.html>.

TOPCAT data may be requested via the Biologic Specimen and Data Repository Information Coordinating Center of the National Heart, Lung, and Blood Institute of the National Institutes of Health (<https://biolinc.nhlbi.nih.gov/studies/topcat/>).

Novartis's data sharing policy is reviewed at <https://www.novartis.com/clinicaltrials/transparency>.

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the manuscript. The manuscript was drafted entirely by the first author and revised with input from all coauthors.

Pre-registered Clinical Trial Number

CHARM-Preserved: NCT00634712; I-PRESERVE: NCT00095238; TOPCAT: NCT00094302; PARAGON-HF: NCT01920711; DELIVER: NCT03619213.

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